

KAIWEN LIU

E-mail: kaiwenliu0811@outlook.com Tel: 07864873860 Address: London N15 3QR Portfolio: <https://kaiwen0418.github.io/about-me>

EDUCATION

Electronic and Information Engineering (MEng)

09.2020 - 06.2024

Imperial College London

Relevant Modules: Computer Vision, Machine Learning, Distributed Algorithm, Digital and Computer Architecture, Discrete Mathematics and Data Structures, Communication Networks, Embedded System

EXPERIENCE

Software Development Engineer

06.2024 - 04.2026

Techblox Limited, London

- Developed backend services for customer-facing products and internal operational tools using Python, TypeScript, FastAPI, and PostgreSQL, supporting transactions processing, wallets managements, external service integrations, and end-to-end status tracking.
- Improved API response times by 20% through PostgreSQL query optimisation and Redis caching, and built BullMQ-based asynchronous workflows to improve reliability and reduce RPC calls and costs.
- Developed more than 50 RESTful API endpoints secured with JWT authentication, role-based access control, request validation, and consistent error handling to support secure, scalable service integration.
- Built unit and integration test suites covering APIs, database operations, queue processing, and transaction workflows, harnessing autonomous AI coding agents with a test-oriented workflow for implementation validation, debugging, and regression verification prior to GitHub Actions and GCP deployment pipelines.

Embedded System Teaching Assistant

02.2024 - 05.2024

Imperial College London, London

- Supported third-year students on an STM32-based music synthesiser project, advising on C++ development, real-time debugging, code optimisation, testing, and hardware integration.

Software Development Intern

06.2023 - 12.2023

SimpleTex Education Technology, Remote

- Built cross-platform AI education applications using Vue.js, Electron.js, and FastAPI, serving more than 200,000 users across mobile, desktop, and web platforms.
- Developed a LaTeX-to-Braille translator supporting China's national mathematical Braille standard, implementing notation rules distinct from Nemeth and Unified English Braille for use by teachers and students at Zhangjiakou Special Education School.

PROJECTS

Agent Benchmarking Platform (TypeScript, Next.js, Redis, PostgreSQL, Docker)

- Built a real-time benchmarking platform for autonomous AI agents, coordinating concurrent browser-based tasks while streaming agent actions, tool calls, replay events, and scoring results to the web interface.
- Architected and deployed a horizontally scalable runtime using stateless Node.js services, partitioned Redis Streams workers, and PostgreSQL-backed transactional state, with dead-letter recovery and Docker-based CI/CD.

Prediction Market Intelligence Dashboard (Python, FastAPI, React.js)

- Built a full-stack prediction market dashboard using FastAPI and React.js, combining WebSocket-based real-time streaming with NumPy analytics to monitor live market activity, price movements, volatility, and aggregated trading signals through REST APIs and real-time data feeds.

.NET Circuit Simulator (F#, .NET, AvaloniaUI)

- Reimplemented a circuit simulator on .NET using F# and an Avalonia MVU interface, reducing package size by 80% and improving memory efficiency by replacing the previous Electron-based implementation.

SKILLS

Languages: **TypeScript, JavaScript, Python, SQL, F#, Verilog**

Technologies: **Node.js, FastAPI, PostgreSQL, Redis, BullMQ, Docker, Next.js, React.js, GitHub Actions**